Pontryagin's Minimum Principle and Linear Regulators

Optimal Control, Homework 7

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology •
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English summary of my original Persian report, which is kept as a PDF in this folder.

The original is handwritten (tablet) in Persian. The problems come from Chapter 5 of Kirk, Optimal Control Theory.

Problem 5-2 (necessary conditions with a bounded input). For a nonlinear second-order system with $\mathcal{H} = x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}u^2 + p_1 x_2 + p_2[(1 - x_1^2)x_2 - x_1 + u]$, I wrote the costate equations $\dot{p}^\star = -\partial\mathcal{H}/\partial x$. With an unconstrained input, $\partial\mathcal{H}/\partial u = 0$ gives $u^\star = -p_2^\star$. With $|u| \le 1$, $u^\star = -p_2^\star$ inside the bound and saturates at ∓ 1 when $|p_2^\star| > 1$.

Problem 5-6 (finite-horizon regulator). For $\dot{x} = x + u$ and $J = \int_0^T (1.5x^2 + 0.5u^2)\,dt$, I solved the state–costate system with the transition matrix $\varphi(t)$ and the condition $p^\star(T) = 0$. This gives a closed-form time-varying feedback $u^\star(t) = -F(t)x^\star(t)$, which tends to $u^\star = -3x$ as $T \to \infty$.

Problem 5-11 (fixed versus free final state). For $\dot{x} = -x + u$ and $J = \int_0^1 \frac{1}{2}(3x^2 + u^2)\,dt$, I derived the optimal feedback for a fixed end state and for a free one. Both are ratios of exponentials in $t_f - t$.

Problem 5-14 (infinite-horizon regulator). For $\dot{x} = ax + u$ with $J = \int_0^\infty (qx^2 + ru^2)\,dt$, the algebraic Riccati equation gives the closed-loop pole $-\sqrt{a^2 + q/r}$, stable for either sign of a. As $q/r \to \infty$ the pole moves to $-\infty$. For a second-order plant I solved the 2×2 Riccati equation. The stabilising root, $K_{12} = 4$, $K_{22} = 2$, places the closed-loop poles at -2 and -3.